

SMART INDIA HACKATHON 2026

CAPACITY CONNECT

A Digital Capacity Building & Learning Management Portal

Problem Statement ID [Official PS ID]	Problem Statement Title CAPACITY CONNECT - A Digital Capacity Building and Learning Management Portal
Theme [Official Theme]	PS Category Software
Team ID [Team ID]	Team Name [Registered Team Name]

LEARN

GROW

CONNECT

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IDEA TITLE & PROPOSED SOLUTION

IDEA TITLE

CAPACITY CONNECT
Smart Digital Platform for Learning, Competency Development & Knowledge Sharing

PROPOSED SOLUTION

A centralized web-based platform connecting employees, trainers, administrators and management for end-to-end organizational learning and capacity building.

KEY FEATURES

- Employee learning, course enrollment and progress tracking
- Trainer course, module, material and quiz management
- Admin control of users, departments, courses and competencies
- Management analytics and training reports
- Competency tracking and skill-gap detection
- AI-based personalized course recommendations
- Certificates and a knowledge-sharing hub

HOW IT ADDRESSES THE PROBLEM

Learning → Assessment → Competency → Skill Gap → Recommendation → Targeted Training → Improvement

INNOVATION / UNIQUENESS

LEARNING	ASSESSMENT	COMPETENCY	SKILL GAP
COURSES + MATERIAL	QUIZ + SCORE	SKILL LEVEL	IDENTIFY NEED
AI RECOMMENDATION	TARGETED TRAINING	PROGRESS	CERTIFICATE

Not just an LMS: CAPACITY CONNECT combines learning, assessment, competency development, skill-gap identification, personalized recommendations, knowledge sharing and analytics in one platform.

TECHNICAL APPROACH

Frontend React.js + Tailwind CSS	Backend Python FastAPI	Database PostgreSQL	Authentication JWT
AI Engine Python / ML recommendation layer	Analytics Interactive charts & reports	Version Control Git + GitHub	Deployment Cloud-ready web architecture

SYSTEM ARCHITECTURE & IMPLEMENTATION FLOW

USER ROLES	WEB APP	BACKEND	DATA + AI	OUTPUT
Employee Trainer Admin Management	React + Tailwind Dashboards & Learning UI	FastAPI REST APIs RBAC	PostgreSQL AI Recommendation Skill Gap	Learning Analytics Certificate

Login → Role Identification → Dashboard → Course Enrollment → Learning Modules → Quiz/Assessment → Progress → Competency Analysis → Skill Gap → AI Recommendation → Training Completion → Certificate

FEASIBILITY & VIABILITY

TECHNICAL FEASIBILITY

- Open-source and widely adopted technologies
- Scalable web architecture
- PostgreSQL for structured learning and user data
- Python supports both backend and AI components
- Cloud-ready deployment

OPERATIONAL FEASIBILITY

- Browser-based and centralized
- Role-based workflows for different users
- Simple dashboard-driven operation
- Can support different organizational departments
- Digital content reduces manual coordination

CHALLENGES → STRATEGIES

- Large training data → scalable database design
- User adoption → simple UI and onboarding
- Data security → authentication + role-based access
- AI accuracy → feedback and performance-based refinement
- Content quality → trainer/admin approval workflow

VIABILITY HIGHLIGHT

The solution can begin as a focused prototype and scale toward organization-wide deployment by expanding users, courses, competencies, content and analytics without changing the core workflow.

IMPACT & BENEFITS

EMPLOYEES Personalized learning • Progress tracking • Skill development • Certificates	TRAINERS Course creation • Digital materials • Assessments • Learner performance
ADMINISTRATORS Centralized management • Users • Courses • Departments • Competencies • Reports	MANAGEMENT Analytics • Department performance • Skill gaps • Training effectiveness • Decisions

OVERALL BENEFITS

Social Better-skilled workforce	Economic Less manual training overhead	Environmental More digital, less paper-based learning	Organizational Centralized data & faster decisions
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Expected outcome: A centralized learning ecosystem that helps organizations train people, measure learning, identify capability gaps and guide the next best learning action.

RESEARCH & REFERENCES

- Smart India Hackathon 2026 Idea Submission Template — used for the required six-slide structure and submission guidance.
- Learning Management Systems — research area for digital course delivery, assessment and progress tracking.
- Competency-Based Learning — research area for competency measurement and development.
- Skill Gap Analysis — research area for comparing current capability with required capability.
- Personalized Learning & AI Recommendation — research area for recommending relevant training from learner data.
- Knowledge Management — research area for organizational knowledge sharing and reusable learning resources.

REFERENCE / RESEARCH LINKS

Source / Area	Purpose
Smart India Hackathon 2026 submission template	Submission format, six-slide limit, content pointers and PDF requirement
Learning Management Systems	Course delivery, assessment and learning progress
Competency & Skill Gap Analysis	Capability measurement and targeted development
AI-based Personalized Learning	Recommendation of relevant learning paths
Knowledge Management	Sharing organizational knowledge and best practices

Note: Replace [Official PS ID], [Official Theme], [Team ID] and [Registered Team Name] with the exact details from the SIH portal before submission.
Template requirement followed: maximum 6 slides, concise points/diagrams, and final upload in PDF format.